PAPER_666: UQFF Gravitational Wave Power Suppression

Subtitle: Derives UQFF suppression factors on GW power (Peters formula) from aether absorption, SCm damping, f_TRZ, and U_m string impedance. **Module:** UQFFGWSuppression
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Abstract

UQFF vacuum fields suppress gravitational wave emission through four mechanisms: aether absorption, superconductive damping, negentropic reversal, and magnetic string impedance. We derive the total P_GW_UQFF and validate against GW150914.

1. Standard GW Power (Peters Formula)

$$P_{GW} = \frac{32}{5}\frac{G^4}{c^5}\frac{m_1^2 m_2^2 (m_1 + m_2)}{r^5}$$

2. UQFF Suppression Factors

S_UA — Aether Absorption

$$S_{UA} = 1 - \frac{\rho_{UA}}{\rho_{crit}}$$

S_SCm — Superconductive Damping

$$S_{SCm} = \exp\left(-\frac{\rho_{SCm}\, r_s}{k_B T_H}\right)$$

S_TRZ — Negentropic

$$S_{TRZ} = 1 - f_{TRZ} = 0.9$$

S_Um — String Impedance

$$S_{U_m} = \exp\left(-\frac{U_m}{\omega_{GW}\, c}\right), \quad \omega_{GW} = c/r_s$$

3. Modified GW Power

$$P_{GW,UQFF} = P_{GW} \cdot S_{UA} \cdot S_{SCm} \cdot S_{TRZ} \cdot S_{U_m}$$

4. Strain Suppression

$$h_{UQFF} = h_{GR}\sqrt{\frac{P_{GW,UQFF}}{P_{GW}}}$$

5. GW150914 Comparison

m1=36 M⊙, m2=29 M⊙, r=350 R⊙: h_GR ≈ 1×10^{-21}; h_UQFF ≈ 0.9×10^{-21} (~10% suppression for dominant S_TRZ).

6. C++ Module

`UQFFGWSuppression.h / .cpp` — Session 172 CP4 #250 — `UQFFGWSuppressionCalculator`

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